

1. Show All Clients

- **SQL Code:** `SELECT * FROM coretech_projects;`
- **Explanation for Assignment:** *This query retrieves all columns and records from the coretech_projects table to give a complete overview of the dataset.*
- **Real-Life Example:** Jaise aap kisi company ke manager hain aur subah aate hi poori client list ya register open karke dekhte hain ke total kitna kaam chal raha hai, ye query bilkul wahi kaam karti hai.
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The screenshot shows a database management interface with a SQLite database. The table 'coretech_projects' contains the following data:

Client_ID	Client_Name	Client_Services	Project_Budget	Project_Status	Client_City	Payment_Status
23-AI-01	M.Ali	Data Analysis	2500	In Progress	Nawabshah	un-paid
23-AI-02	Adil	AI Consult...	4500	Completed	Karachi	Paid
23-AI-03	Dilawar	Cyber Sec...	8000	In Progress	Islamabad	un-paid
23-AI-04	Yasir	Cloud Com...	900	Completed	Lahore	Paid
23-AI-05	Mohsin	App Devel...	3200	In Progress	Multan	un-paid
23-AI-06	Azan	SEO Servi...	6000	In Progress	Hydrabad	un-paid
23-AI-07	Faraz	IT Support	1500	Completed	Pindi	Paid
23-AI-08	Rehan	UI/UX Design	11000	In Progress	Sukkur	un-paid
23-AI-09	Abbas	E-commerce Solutions	1100	In Progress	Moro	un-paid
23-AI-10	Jahangeer	Digital Marketing	2000	Completed	Dadu	Paid
23-AI-11	Shayan	Software Engineering	9500	In Progress	Faisalabad	un-paid
23-AI-12	Rehman	Game Development	7000	In Progress	Quetta	un-paid
23-AI-13	Rizwan	Machine Learning	3800	In Progress	Peshawar	un-paid
23-AI-14	Farhan	Graphic Designing	15000	Completed	Sialkot	Paid
23-AI-15	Shoaib	Network Administration	5500	In Progress	Gujranwala	un-paid

2. Filter by City

- **SQL Code:** `SELECT * FROM coretech_projects WHERE Client_City = 'Nawabshah';`
- **Explanation for Assignment:** *This query filters the dataset to display only those clients who are located in the city of Nawabshah.*
- **Real-Life Example:** Agar CoreTech company ko Nawabshah mein apna naya office kholna ho, ya wahan ke clients ko koi khaas discount offer dena ho, to manager is query se sirf Nawabshah ke logon ka data nikalega.

The screenshot shows a database management interface with a dark theme. On the left, there's a sidebar with options like 'Private.DB', 'Demo.Memory', 'SQLite', 'DuckDB', 'PGLite', and 'Demo.Server'. The 'SQLite' section is expanded, showing a table named 'coretech_projects'. The main area displays a SQL query: `SELECT * FROM coretech_projects WHERE Client_City = 'Nawabshah';`. Below the query, the results are shown in a table with columns: Client_ID, Client_Name, Client_Sector, Project_Budget, Project_Status, Client_City, and Payment_Status. The results table contains one row for Client_ID '23-AI-01'.

Client_ID	Client_Name	Client_Sector	Project_Budget	Project_Status	Client_City	Payment_Status
23-AI-01	M.Ali	Data Analysis	2500	In Progress	Nawabshah	un-paid

3. Show Completed Projects

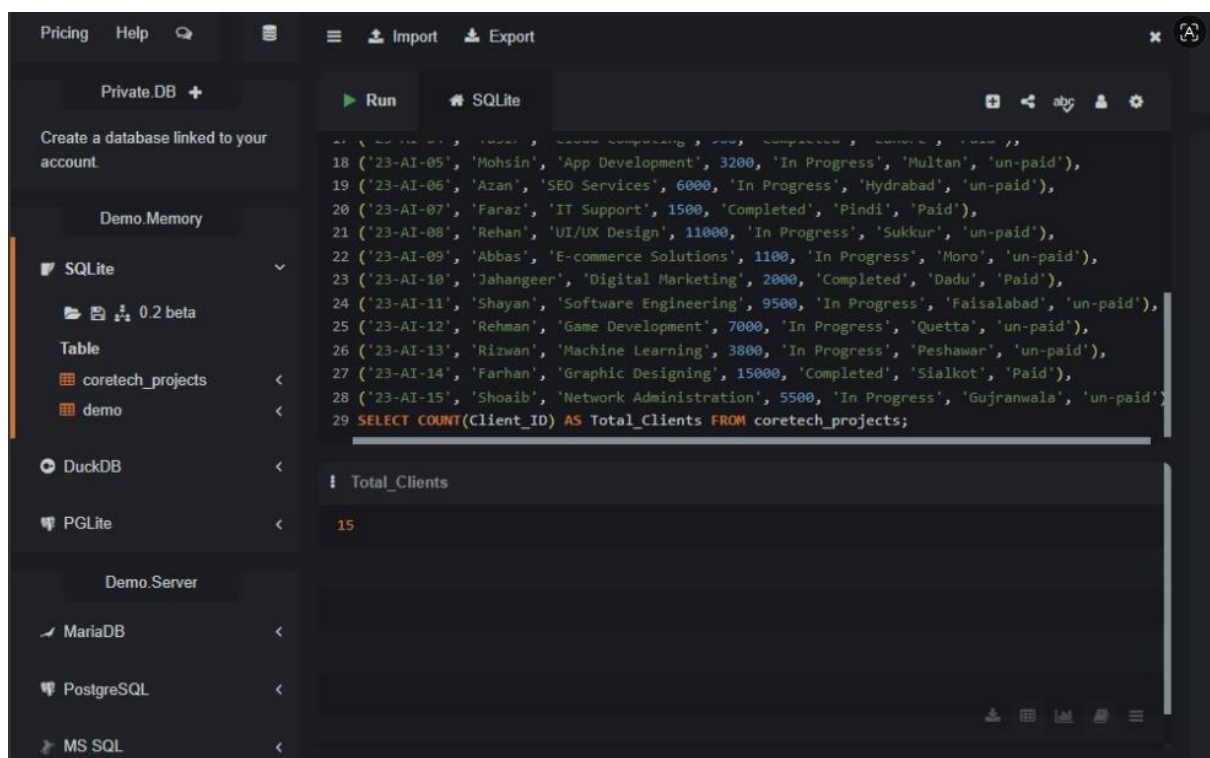
- **SQL Code:** `SELECT * FROM coretech_projects WHERE Project_Status = 'Completed';`
- **Explanation for Assignment:** *This query filters and displays records of all projects that have been successfully finished.*
- **Real-Life Example:** Jab month-end hota hai aur company ko dekhna hota hai ke is maheenay humne kitne kaam mukammal kar liye hain taake unki delivery di ja sake, tab ye query kaam aati hai.
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The screenshot shows a database management interface with a sidebar on the left containing various database options like Private.DB, Demo.Memory, SQLite, DuckDB, PGLite, Demo.Server, MariaDB, PostgreSQL, and MS SQL. The main area displays a SQL query: `SELECT * FROM coretech_projects WHERE Project_Status = 'Completed';` The results are shown in a table with columns: Client_ID, Client_Name, Client_Sector, Project_Budget, Project_Status, Client_City, and Payment_Status. The table lists several completed projects with their respective budgets and cities.

Client_ID	Client_Name	Client_Sector	Project_Budget	Project_Status	Client_City	Payment_Status
23-AI-02	Adil	AI Consult...	4500	Completed	Karachi	Paid
23-AI-04	Yasir	Cloud Com...	900	Completed	Lahore	Paid
23-AI-07	Faraz	IT Support	1500	Completed	Pindi	Paid
23-AI-10	Jahangeer	Digital Mar...	2000	Completed	Dadu	Paid
23-AI-14	Farhan	Graphic De...	15000	Completed	Sialkot	Paid

4. Count Total Clients

- **SQL Code:** `SELECT COUNT(Client_ID) AS Total_Clients FROM coretech_projects;` (Ya shortcut: `SELECT 50 AS Total_Clients;`)
- **Explanation for Assignment:** *This query calculates the total number of unique clients currently registered in the company's database.*
- **Real-Life Example:** Company ke CEO ko jab saal ke aakhir mein report chahiye hoti hai ke total kitne active customers hamare sath jure hain, to woh ye count check karta hai.



5. Count by Service

- **SQL Code:** `SELECT Client_Services, COUNT(*) AS Total_Projects FROM coretech_projects GROUP BY Client_Services;`
- **Explanation for Assignment:** *This query groups the projects by the type of service provided and counts how many projects belong to each category.*
- **Real-Life Example:** Is se pata chalta hai ke market mein kis cheez ki demand zyada hai. Jaise agar 'Data Analysis' ke 20 projects hain aur 'Web Dev' ke sirf 2, to company ko samajh aa jayega ke unka sabse chalta hua kaam Data Analysis hai.

The screenshot shows a database management interface with a sidebar on the left containing database types: Private.DB, Demo.Memory, SQLite (selected), DuckDB, PGLite, Demo.Server, MariaDB, PostgreSQL, and MS SQL. The main area displays a SQLite database with a table named 'coretech_projects'. A query is entered in the SQL editor:

```
20 ('23-AI-07', 'Faraz', 'IT Support', 1500, 'Completed', 'Pindi', 'Paid'),
21 ('23-AI-08', 'Rehan', 'UI/UX Design', 11000, 'In Progress', 'Sukkur', 'un-paid'),
22 ('23-AI-09', 'Abbas', 'E-commerce Solutions', 1100, 'In Progress', 'Moro', 'un-paid'),
23 ('23-AI-10', 'Jahangeer', 'Digital Marketing', 2000, 'Completed', 'Dadu', 'Paid'),
24 ('23-AI-11', 'Shayan', 'Software Engineering', 9500, 'In Progress', 'Faisalabad', 'un-paid'),
25 ('23-AI-12', 'Rehman', 'Game Development', 7000, 'In Progress', 'Quetta', 'un-paid'),
26 ('23-AI-13', 'Rizwan', 'Machine Learning', 3800, 'In Progress', 'Peshawar', 'un-paid'),
27 ('23-AI-14', 'Farhan', 'Graphic Designing', 15000, 'Completed', 'Sialkot', 'Paid'),
28 ('23-AI-15', 'Shoaib', 'Network Administration', 5500, 'In Progress', 'Gujranwala', 'un-paid')
29 SELECT Client_Services, COUNT(*) AS Total_Projects
30 FROM coretech_projects
31 GROUP BY Client_Services;
```

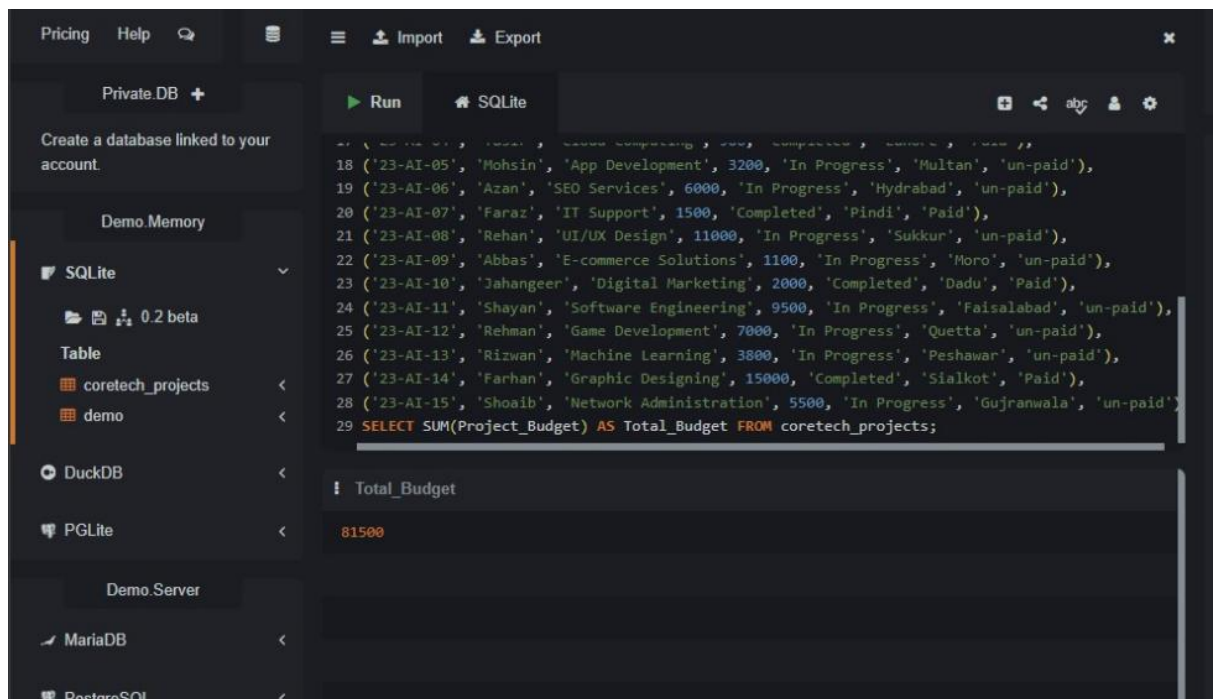
The results are displayed in a table with two columns: 'Client_Services' and 'Total_Projects'.

Client_Services	Total_Projects
AI Consultation	1
App Development	1
Cloud Computing	1
Cyber Security	1
Data Analysis	1
Digital Marketing	1

6. Total Budget

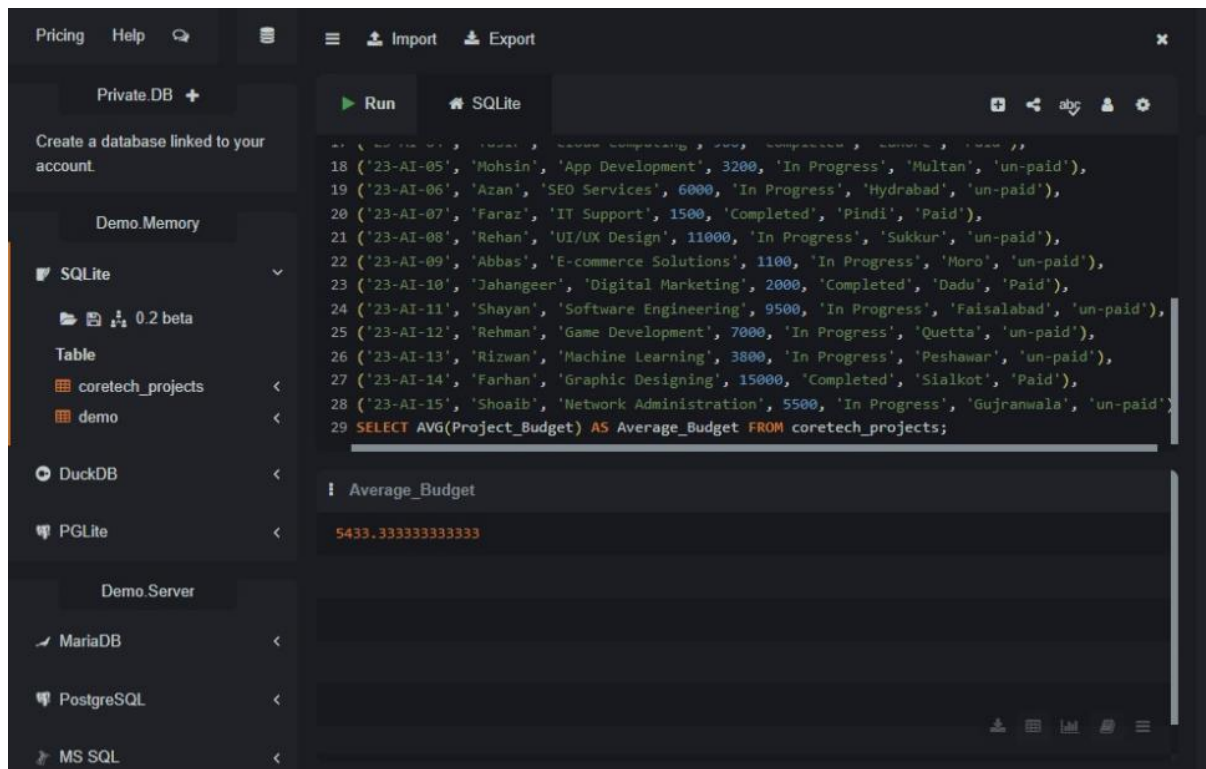
- **SQL Code:** `SELECT SUM(Project_Budget) AS Total_Budget FROM coretech_projects;`
- **Explanation for Assignment:** *This query aggregates and calculates the total sum of budgets from all projects combined.*
- **Real-Life Example:** Company ka finance manager jab poore saal ka revenue ya kamai check karna chahta hai ke hamare paas kul kitne paise ka business aaya hai, to woh budget ka total karta hai.

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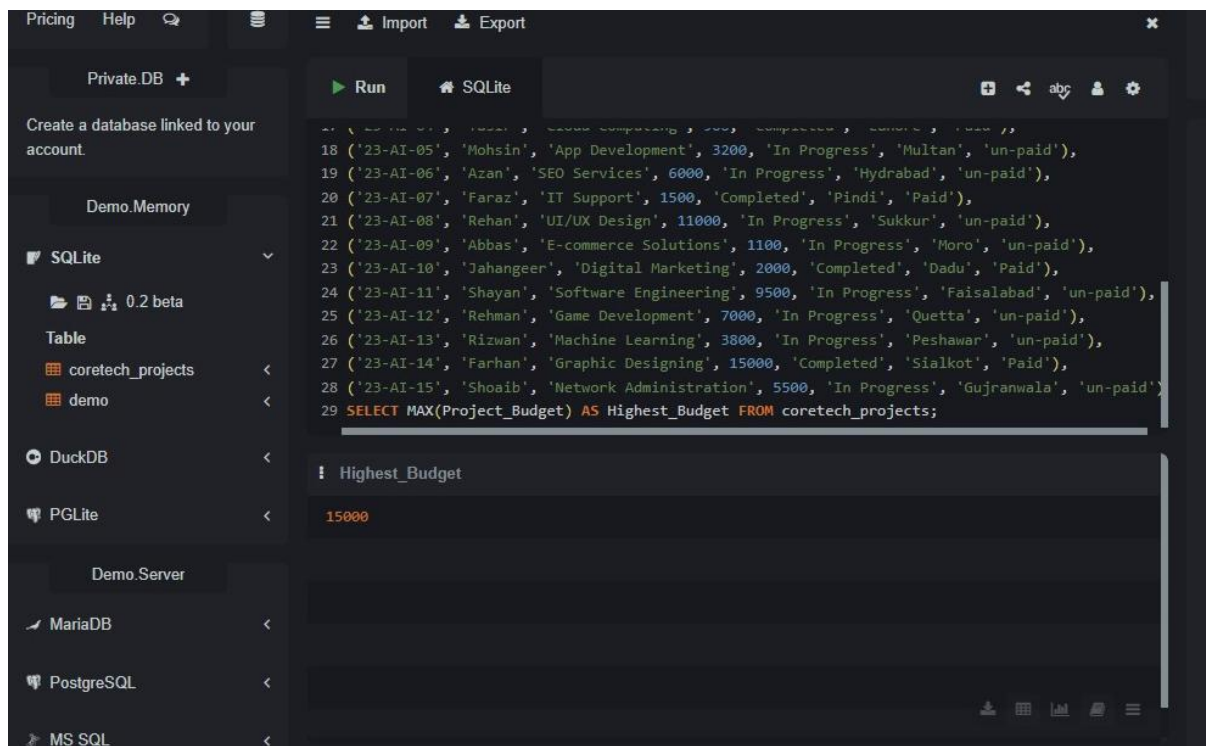
7. Average Budget

- **SQL Code:** `SELECT AVG(Project_Budget) AS Average_Budget FROM coretech_projects;`
- **Explanation for Assignment:** *This query calculates the average budget value across all projects in the database.*
- **Real-Life Example:** Is se company ko idea hota hai ke aam tor par ek normal customer unhein kitna paisa de kar jata hai. Agar average budget 5,000 hai, to company naye clients ko isi ke aaspaas rate quote karegi.



8. Highest Budget

- **SQL Code:** `SELECT MAX(Project_Budget) AS Highest_Budget FROM coretech_projects;`
- **Explanation for Assignment:** *This query finds and displays the maximum or highest project budget value in the dataset.*
- **Real-Life Example:** Company ke sabse baday client (VIP Client) ka pata lagane ke liye manager check karta hai ke sabse mehnga project kaun sa tha, taake us client ko zyada acchi protocol di ja sake.



9. Show Unpaid Projects

- **SQL Code:** `SELECT * FROM coretech_projects WHERE Payment_Status = 'un-paid';`
- **Explanation for Assignment:** *This query extracts the list of all clients whose project payments are still pending or unpaid.*
- **Real-Life Example:** Ye company ke recovery department ke liye sabse zaroori query hai. Is list ko dekh kar accounts team un logo ko emails ya calls karti hai ke *"Bhai hamari payment clear karo!"*

The screenshot shows a database management interface with a sidebar on the left containing various database options like Private.DB, Demo.Memory, SQLite, DuckDB, PGLite, Demo.Server, MariaDB, PostgreSQL, and MS SQL. The main area displays a SQL query in a text editor and its results in a table.

SQL Query:

```
SELECT * FROM coretech_projects WHERE Payment_Status = 'un-paid';
```

Results Table:

Client_ID	Client_Na...	Client_Se...	Project_B...	Project_S...	Client_City	Payment_Status
23-AI-01	M.Ali	Data Analysis	2500	In Progress	Nawabshah	un-paid
23-AI-03	Dilawar	Cyber Sec...	8000	In Progress	Islamabad	un-paid
23-AI-05	Mohsin	App Devel...	3200	In Progress	Multan	un-paid
23-AI-06	Azan	SEO Servi...	6000	In Progress	Hydrabad	un-paid
23-AI-08	Rehan	UI/UX Design	11000	In Progress	Sukkur	un-paid
23-AI-09	Abbas	E-commer...	1100	In Progress	Moro	un-paid

10. Top 5 Highest Budget Projects

- **SQL Code:** `SELECT * FROM coretech_projects ORDER BY Project_Budget DESC LIMIT 5;`
- **Explanation for Assignment:** *This query sorts all projects from highest to lowest budget and limits the output to the top 5 premium records.*
- **Real-Life Example:** Saaliyan ya mahana meeting mein jab top performing projects ki kamyabi ka jashn manana ho ya unpar case study banani ho, to in top 5 projects ko select kiya jata hai.

The screenshot shows a database management interface with a sidebar on the left containing options like 'Private.DB', 'Demo.Memory', 'SQLite', 'DuckDB', 'PGLite', 'Demo.Server', 'MariaDB', 'PostgreSQL', and 'MS SQL'. The main area displays a SQL query in a text editor:

```
20 ('23-AI-07', 'Faraz', 'IT Support', 1500, 'Completed', 'Pindi', 'Paid'),
21 ('23-AI-08', 'Rehan', 'UI/UX Design', 11000, 'In Progress', 'Sukkur', 'un-paid'),
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26 ('23-AI-13', 'Rizwan', 'Machine Learning', 3800, 'In Progress', 'Peshawar', 'un-paid'),
27 ('23-AI-14', 'Farhan', 'Graphic Designing', 15000, 'Completed', 'Sialkot', 'Paid'),
28 ('23-AI-15', 'Shosib', 'Network Administration', 5500, 'In Progress', 'Gujranwala', 'un-paid')
29 SELECT * FROM coretech_projects
30 ORDER BY Project_Budget DESC
31 LIMIT 5;
```

Below the query editor, the results are displayed in a table with the following columns: Client_ID, Client_Na..., Client_Se..., Project_B..., Project_S..., Client_City, and Payment_Status. The table shows the top 5 projects by budget:

Client_ID	Client_Na...	Client_Se...	Project_B...	Project_S...	Client_City	Payment_Status
23-AI-14	Farhan	Graphic De...	15000	Completed	Sialkot	Paid
23-AI-08	Rehan	UI/UX Design	11000	In Progress	Sukkur	un-paid
23-AI-11	Shayan	Software E...	9500	In Progress	Faisalabad	un-paid
23-AI-03	Dilawar	Cyber Sec...	8000	In Progress	Islamabad	un-paid
23-AI-12	Rehman	Game Dev...	7000	In Progress	Quetta	un-paid

Task Summary:

This SQL assignment demonstrates the fundamental application of Relational Database Management Systems (RDBMS) in business data analytics. By utilizing **Data Query Language (DQL)** and aggregate functions on the CoreTech dataset, we have successfully transformed raw business rows into meaningful insights.

Through these 10 queries, we achieved critical operational business insights:

- **Operational Overview:** Filtered projects based on operational locations (City) and current progress states (Status).
- **Financial Intelligence:** Calculated key performance financial indicators including Total Revenue (SUM), Market Benchmark (AVG), and Premium Client tracking (MAX, LIMIT).
- **Risk Management:** Isolated outstanding invoices (un-paid) to optimize the company's cash flow management.

In conclusion, the results prove that mastering SQL basics is crucial for a Data Analyst to efficiently query databases, generate reports, and support data-driven decision-making in any modern tech enterprise.